

AUTOMATA

Syllabus of TOC

- ① Regular Language and Finite Automata (80%)
- ② Context free Language and push down Automata (10%)
- ③ REC, RE and Turing Machine (10%)

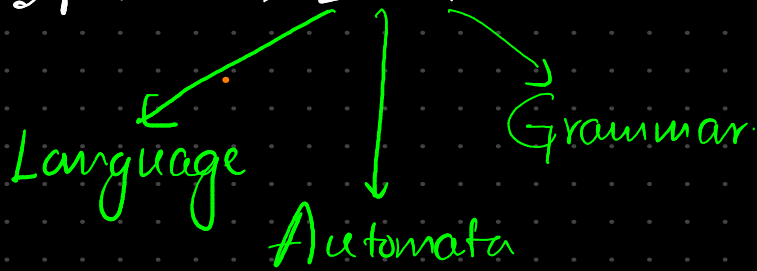
④ Closure properties and undecidability ★★

├ Language
├ Grammar
└ Automata (Machine)

├ Tables
├ language identify
└ Automata

TOC → concept → principles.

3 pillars → LAG



Symbol.

$\{a, b, c, \dots, 0, 1, 2, \dots\}$

String → Collection of alphabet, sequence.

Alphabet $\Sigma(a, b)$
Finite Set of symbol.

Language → Collection of strings.

$\Sigma(a, b)$ $\Rightarrow L_1 = \text{string of length 3}$

$\Rightarrow L_2 = \infty$

All the strings start with 'a' and end with 'a'.

$\{a a a, a a b, a b a, a b b, b a a, b a b, b b a, b b b\}$

$\{a, a a, a a a, a b b a, a b c b a, \dots, \infty\}$

Language length strings '0'
 $\Sigma^0 = \epsilon$

AUTOMATA

Language

→ FA
→ PDA
→ LBA
→ TM

Finite ✓

Infinite

Automata (Model, Machine)

[whether a string is a part of language or not]

Mathematical Abstract Machine

POWER OF SIGMA

Power of Σ $\Sigma(a,b)$

Σ^0 = Set of all strings with length 0 = λ, ϵ

Σ^1 = Set of all strings with length 1 = $\{a, b\}$

Σ^2 = Set of all strings with length 2 = $\{aa, ab, ba, bb\}$

Σ^* = (Kleen closure) = ∞ Language
 $\hookrightarrow (a+b)^*$

$\Sigma^+ \rightarrow$ (positive closure) $\rightarrow$ all except ϵ

$$\boxed{\Sigma^* = \Sigma^+ - \Sigma^0}$$

$$\boxed{\Sigma^* - \epsilon = \Sigma^+}$$

GRAMMAR (G):

$\hookrightarrow$ defined as quadruple

$$G = \{V, T, P, S\}$$

Variable

Terminal

Production Rule

Start

$$S \rightarrow aSb \mid \epsilon$$

$\epsilon, aSb, aaSbb, aaaSbbb$

$\epsilon, ab, a^2Sb^2, a^3Sb^3$
 a^2b^2, a^3b^3

a^nSb^n

$a^n b^n$

$n \geq 0$

$$S \rightarrow SS$$

$$S \rightarrow aSb$$

$$S \rightarrow bSa$$

$$S \rightarrow \epsilon$$

ϵ, aSb, bSa

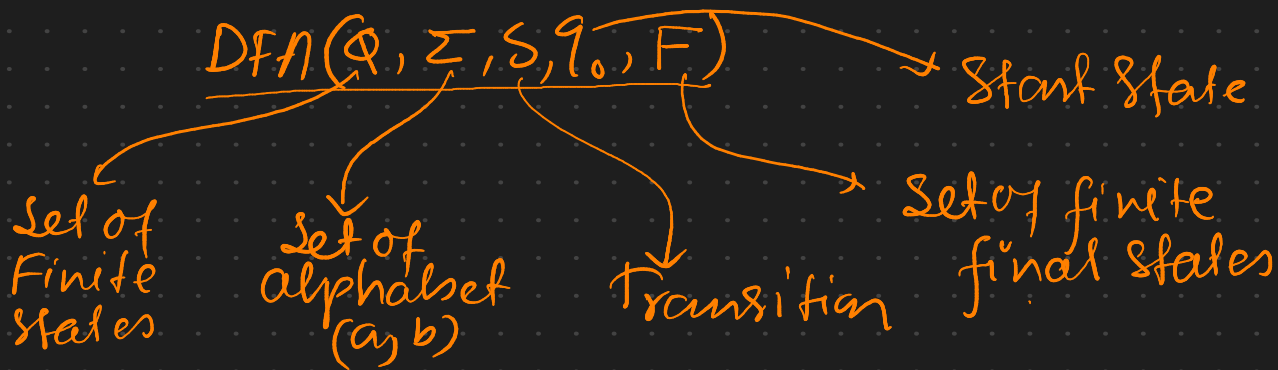
$abba$

Language $\rightarrow$

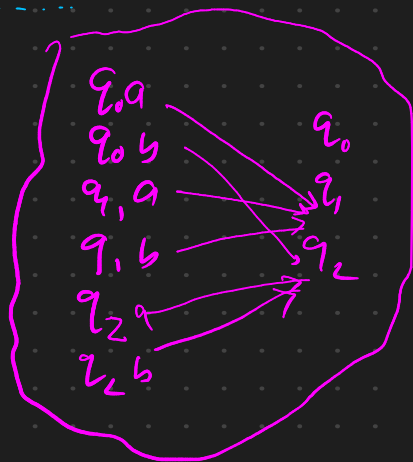
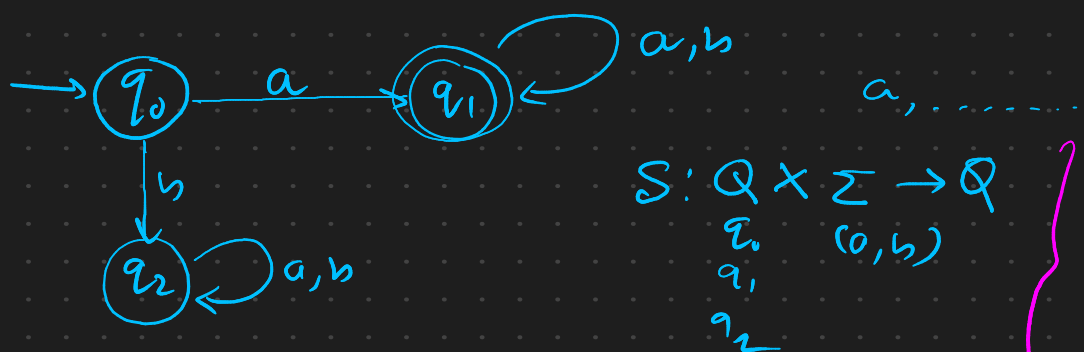
$$\boxed{n_a(w) = n_b(w)}$$

DFA (Deterministic Finite Automata)

$DFA(Q, \Sigma, S, q_0, F)$



String starting with 'a' $\rightarrow$ Language
 $\{a, aa, aaa, ab, \underline{ba}, \dots\}$



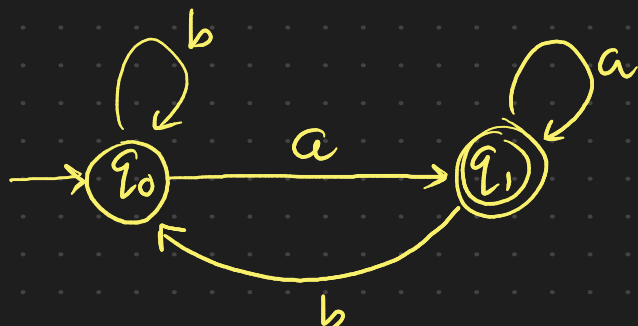
Q. Construct a DFA which accept a lang. of all strings containing 'a'

Ans: $\{a, aa, aba, ba, ab, \underline{b}, \dots\}$



Q. Construct a DFA which accept a language of all strings end with 'a'.

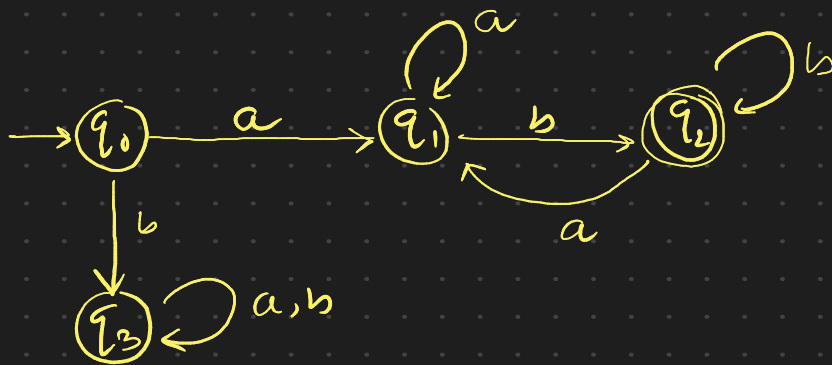
Ans: $L = \{a, aa, aaa, ba, baba, abba, baaba, \underline{ab}, abab, \dots\}$



Q. Construct a DFA which accept a language of all strings starting with 'a' and ending with 'b'.

Ans

$$L = \{ab, abab, ababab, \frac{ba}{x}, \frac{a}{x}, \frac{b}{x}, \dots, \infty\}$$



Q. Construct a DFA which accept a language of all strings not starting with 'a' or not ending with 'b'.

Ans

$$L = \{\lambda, \epsilon, \dots\}$$

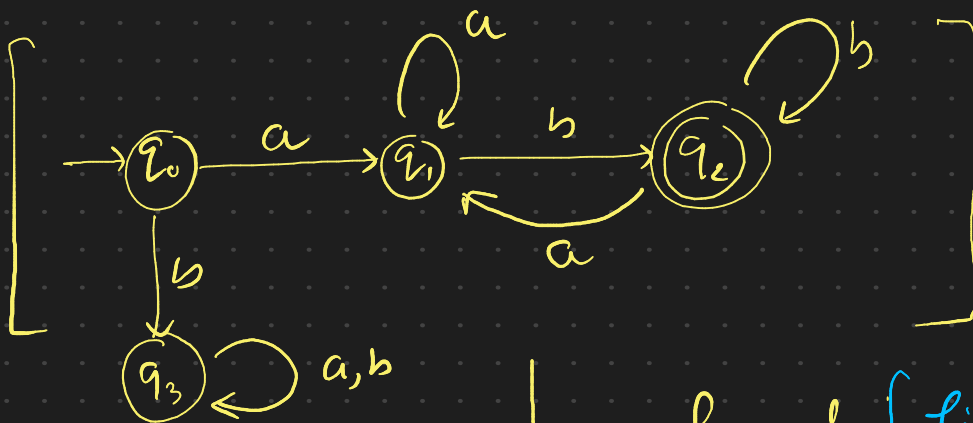
A or B

$$(A \cup B)^c = A^c \cap B^c \quad (A \cup B)$$

Starting with a

(8)

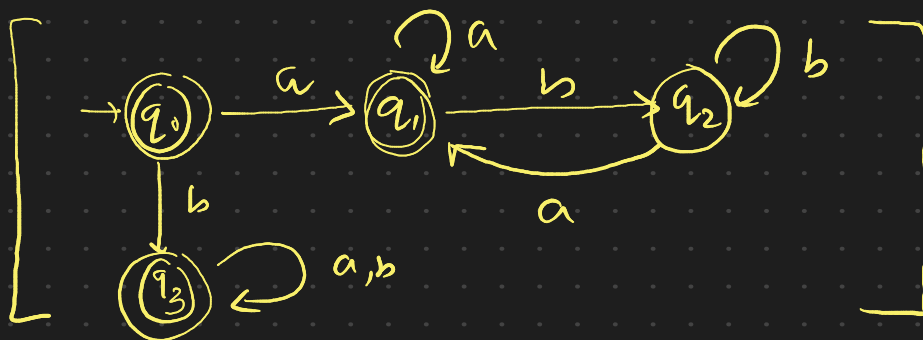
Ending with b



$(A \cup B)$

complement

final $\rightarrow$ non final
non final $\rightarrow$ final



$(A \cup B)^c$

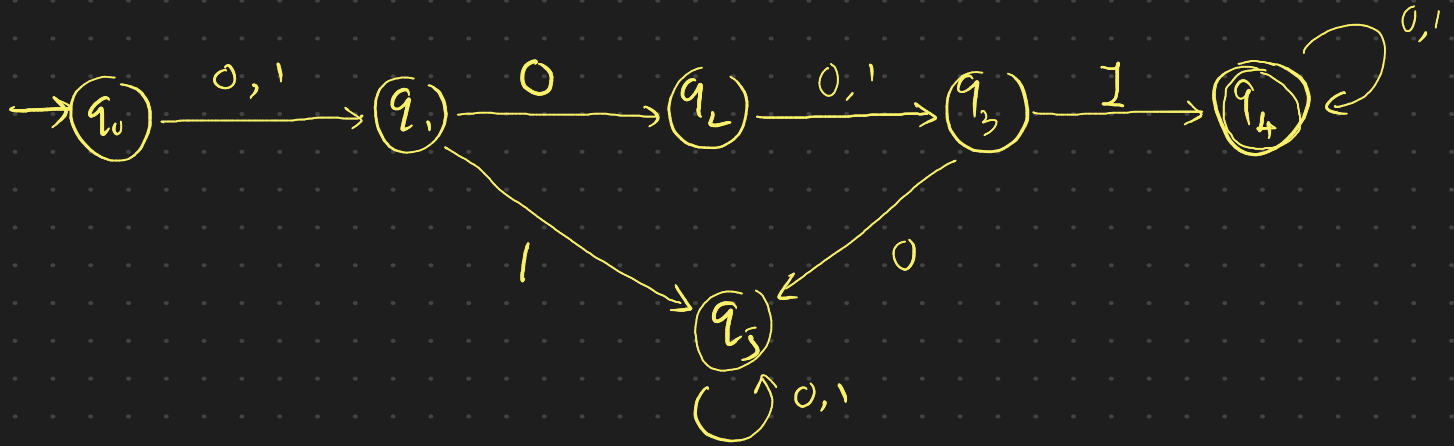
Q. Design a DFA which accepts all strings over (0,1) in which 2nd symbol is 0 and 4th is 1.

Ans.

0 1 0 1 (4)

min.

no. of states = $n+1 = 4+1 = 5$ ✓ (minimum length)



DFA

NDFA/NFA

① deal w/ not allowed

① allowed

② Multiple choices w/ attrib corresponding to an ip.

② available

③ ∈ not allowed

③ ∈ allowed

④ Digital computer

④ Non deterministic feature is not associated with real machines.

⑤ Designing & understanding difficult

⑤ Easy

⑥ No. of states ↑

⑥ ↓